

KJSIT Oct 2026 Hackathon Problem Statements

PS 1: Title: Sell Smart: "Where and When to Sell" Advisory for Farmers and FPOs

Domain: Agriculture

Background

Most small farmers sell their produce at the nearest mandi right after harvest, when heavy arrivals push prices down. A mandi 40 km away may be paying far more, and prices may rise in two weeks, but farmers have no easy way to know this or to judge whether waiting is worth the risk. Existing tools mostly show today's prices or give a single price forecast. They don't answer the real decision a farmer faces: *should I sell now or hold, and which mandi gives me the most money in hand?*

Problem

Farmers and Farmer Producer Organisations (FPOs) lack a decision tool that combines future prices, transport costs, and crop spoilage into one clear recommendation. As a result, they make distress sales, lose income to intermediaries, and either wait too long and suffer spoilage or sell too early and miss better prices.

Proposed solution

Build an advisory system that tells farmers where and when to sell for the best net return.

- Net price comparison across mandis: Forecast prices at 10–15 nearby mandis for the next 1–3 weeks and subtract estimated transport cost from the farmer's village, so the recommendation reflects actual money in hand, not just the highest mandi price.
- Sell now or hold advice: Weigh expected price gain against spoilage and storage loss for each crop. For example, onions can be held for weeks, while tomatoes must be sold within days. The output is a clear recommendation such as "Hold 10 days, sell at Lasalgaon, expected gain ₹180/quintal."
- Honest uncertainty: Show a likely price range with a confidence level such as high, medium, or low instead of a single number, so farmers understand the risk of waiting.
- WhatsApp and voice interface: Farmers send their crop, quantity, and village by WhatsApp text or voice note in Marathi or Hindi, and receive a short reply in their language, with no app download needed.
- Proven rupee impact: Backtest the advisory on past seasons to show how much more a farmer would have earned by following it compared with selling at the nearest mandi on harvest day.
- FPO bulk mode: A simple dashboard for FPOs to plan where to send bulk produce across mandis.

Scope for the prototype

One state (for example, Maharashtra), three crops with different shelf lives (onion, tomato, soybean), and 10–15 major mandis.

PS 2 : AI-Powered YouTube Thumbnail Attention Heatmap

Domain: AI & Machine Learning

Problem Statement:

YouTube thumbnails play a critical role in attracting viewers, but creators typically have limited visibility into which parts of a thumbnail are likely to capture a viewer's attention first.

Design and develop an AI-powered system that analyzes a YouTube thumbnail and generates an attention heatmap showing the areas that are most likely to attract visual attention.

The system should go beyond simply identifying objects. It should analyze the composition, visual hierarchy, text placement, faces, subjects, colors, contrast, and other visual elements and estimate how viewer attention may be distributed across the thumbnail.

PS 3 : AI-Powered B-roll Search & Retrieval System

Domain: AI & Machine Learning

Problem Statement:

Video creators spend significant time searching for suitable **B-roll footage** to support their scripts, voiceovers, and primary footage.

Traditional video search relies heavily on filenames, tags, or manually added keywords, making it difficult to find footage when the creator's description does not exactly match the metadata.

Build an AI-powered **semantic B-roll search and retrieval system** that understands the meaning of a user's text/script and retrieves the most relevant video clips from a large video library.

The system should understand **objects, actions, scenes, environments, people, and context**, rather than relying only on keyword matching.

PS 4 : "Is That Really Them?" Deepfake Scam Ad Detector

Domain: Cybersecurity & Digital Trust (Deepfake & Synthetic Media Detection)

Background

Scam ads on YouTube, Instagram and Facebook increasingly use AI-generated videos of well-known creators, business leaders and news anchors. These videos promote fake investment schemes, trading apps and "government-backed" offers. The clips are short, often in Hindi or regional languages, and look convincing on a phone screen. Viewers lose money, and the impersonated creators lose trust with their audience. Most existing deepfake detectors are trained on English, studio-quality face-swap datasets. They return a bare "fake/real" score with no explanation, which isn't useful to an ordinary viewer or to a platform moderator.

Problem

Design an AI system that analyses a short video ad and estimates whether it contains synthetic or manipulated media of a real person. It should explain *why* in terms a non-expert can understand.

Expected features

Audio-visual analysis: Detect lip-sync mismatch, voice cloning artifacts, unnatural blinking or facial boundaries, and inconsistencies between the audio and the video.

Scam signal detection: Flag the typical content patterns of fraud ads, such as guaranteed returns, urgency, requests to move to WhatsApp or Telegram, and fake news-channel branding.

Explainable output: Show a risk level (high, medium or low) along with the specific frames and audio segments that triggered it. For example: "Mouth movement doesn't match speech at 0:07–0:12."

Easy access: Users can paste a link or forward a clip, for example through a WhatsApp bot or a browser extension, and get a result in under a minute.

Scope for the prototype

Clips of 15–90 seconds, in Hindi and English. Teams may use public deepfake datasets plus self-generated samples, clearly labelled.

PS 5 : Retention Predictor, Find the Drop-Off Before You Publish

Domain: AI & Machine Learning (Predictive Analytics / NLP)

Background

Audience retention is one of the strongest signals YouTube uses to recommend a video. Creators only see where viewers dropped off *after* publishing, when it's too late to fix the video. Many drop-offs have predictable causes: a slow intro, a long tangent, a delayed payoff to the title's promise, or repeated information. A good editor spots these by instinct, but most creators don't have one.

Problem

Build an AI system that takes a video's script or transcript, and optionally the video itself, and predicts where viewers are likely to lose interest. It should suggest specific fixes.

Expected features

Predicted retention curve: Estimate a second-by-second or segment-level retention curve for the video.

Reasoned flags: Mark risky segments with reasons, such as "Title promise not addressed until 2:40," "Intro runs 45 seconds before the hook," or "This section repeats 1:10–1:30."

Actionable edits: Suggest concrete changes, such as moving a segment, cutting a section, or adding a hook line, and show the expected improvement.

Validation: Compare the predictions with actual retention data from public or team-provided videos to show accuracy.

Scope for the prototype

English and Hindi videos of 5–15 minutes in 2–3 content categories, such as tech reviews, education and vlogs.